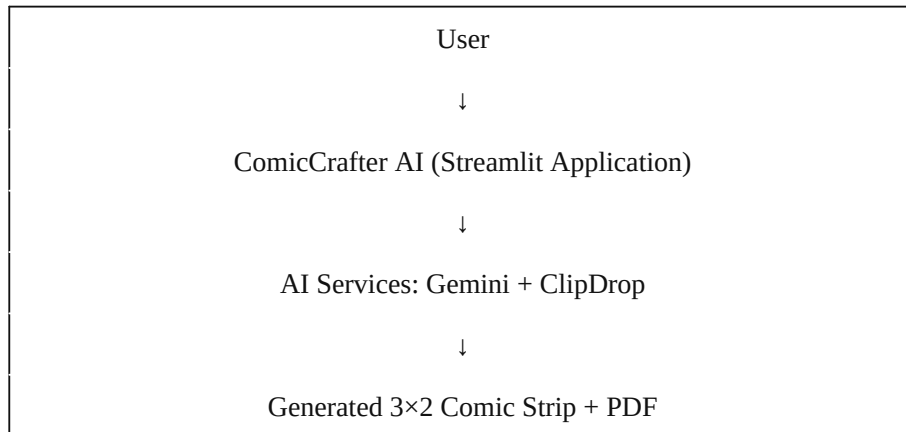


Data Flow Diagram

Date 03 October 2026
Team ID COMICCRAFT-TEAM
Project Name ComicCrafter AI – AI-Powered Comic Strip Generator
Maximum Marks 2 Marks

Level 0 – Context Diagram



Level 1 – Detailed Data Flow

Process	Input	Processing	Output
1. Collect Input	Story prompt + art style	Validate non-empty prompt and allowed style.	Validated request
2. Generate Panels	Scenario + style	Gemini 1.5 Flash creates six descriptions and dialogue; parser extracts panel data.	Six structured panels
3. Generate Images	Panel descriptions + style	ClipDrop text-to-image API creates one image per panel.	Six panel images
4. Process Comic	Six images + six text values	Pillow adds text areas and combines panels into a 3x2 grid.	Final comic PNG
5. Export	Final comic PNG	ReportLab places the image into an A4 PDF.	Downloadable PDF

Data Flow Summary

User → Streamlit → Gemini panel generation → panel parsing → ClipDrop image generation → Pillow text overlay and 3×2 assembly → Streamlit preview → ReportLab PDF download.

External Services

- Gemini API supplies structured panel descriptions and dialogue.
- ClipDrop API converts visual panel descriptions into images.
- No persistent user database is required by the supplied implementation; generated files are stored locally in PANEL_IMAGES/ and OUTPUT/.